Evidencia CRUD

Dentro de mongos podemos ejecutar los cuatros operaciones del CRUD sin ningún problema. Lo interesante de esto es que, aparentemente nada a cambiado, pero en realidad por detrás mongos determina a que shard debe enviar el documento y de donde leerlo en caso de las consultas.

Escribir

```
[direct: mongos] dbemp> db.emp.insertMany([
|   { empno: 7369, ename: "SMITH", job: "CLERK", sal: 800.0, dept: { deptno: 20, dname: "RESEARCH", loc: "DALLAS" } },
|   { empno: 7499, ename: "ALLEN", job: "SALESMAN", sal: 1600.0, dept: { deptno: 30, dname: "SALES", loc: "CHICAGO" } },
|   { empno: 7521, ename: "WARD", job: "SALESMAN", sal: 1250.0, dept: { deptno: 30, dname: "SALES", loc: "CHICAGO" } },
|   { empno: 7566, ename: "JONES", job: "MANAGER", sal: 2975.0, dept: { deptno: 20, dname: "RESEARCH", loc: "DALLAS" } },
|   { empno: 7654, ename: "MARTIN", job: "SALESMAN", sal: 1250.0, dept: { deptno: 30, dname: "SALES", loc: "CHICAGO" } },
|   { empno: 7698, ename: "BLAKE", job: "MANAGER", sal: 2850.0, dept: { deptno: 30, dname: "SALES", loc: "CHICAGO" } },
|   { empno: 7782, ename: "CLARK", job: "MANAGER", sal: 2450.0, dept: { deptno: 10, dname: "ACCOUNTING", loc: "NEW YORK" } },
|   { empno: 7788, ename: "SCOTT", job: "ANALYST", sal: 3000.0, dept: { deptno: 20, dname: "RESEARCH", loc: "DALLAS" } },
|   { empno: 7839, ename: "KING", job: "PRESIDENT", sal: 5000.0, dept: { deptno: 10, dname: "ACCOUNTING", loc: "NEW YORK" } },
|   { empno: 7844, ename: "TURNER", job: "SALESMAN", sal: 1500.0, dept: { deptno: 30, dname: "SALES", loc: "CHICAGO" } },
|   { empno: 7876, ename: "ADAMS", job: "CLERK", sal: 1100.0, dept: { deptno: 20, dname: "RESEARCH", loc: "DALLAS" } },
|   { empno: 7900, ename: "JAMES", job: "CLERK", sal: 950.0, dept: { deptno: 30, dname: "SALES", loc: "CHICAGO" } },
|   { empno: 7902, ename: "FORD", job: "ANALYST", sal: 3000.0, dept: { deptno: 20, dname: "RESEARCH", loc: "DALLAS" } },
|   { empno: 7934, ename: "MILLER", job: "CLERK", sal: 1300.0, dept: { deptno: 10, dname: "ACCOUNTING", loc: "NEW YORK" } }
| ])
```

Leer

```
[direct: mongos] dbemp> db.emp.find({empno: 7369}, )
[
  {
    _id: ObjectId('6ae2cbc042e4712c6fde208c'),
    empno: 7369,
    ename: 'SMITH',
    job: 'CLERK',
    sal: 800,
    dept: { deptno: 20, dname: 'RESEARCH', loc: 'DALLAS' }
  }
]
```

Actualizar

```
[direct: mongos] dbemp> db.emp.updateOne({empno: 7369}, { $set: {sal: 1200}} )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
[direct: mongos] dbemp> db.emp.find({empno: 7369})
[
  {
    _id: ObjectId('6ae2cbc042e4712c6fde208c'),
    empno: 7369,
    ename: 'SMITH',
    job: 'CLERK',
    sal: 1200,
    dept: { deptno: 20, dname: 'RESEARCH', loc: 'DALLAS' }
  }
]
```

Eliminar

```
[direct: mongos] dbemp> db.emp.deleteOne({empno: 7369})
{ acknowledged: true, deletedCount: 1 }
[direct: mongos] dbemp>
```